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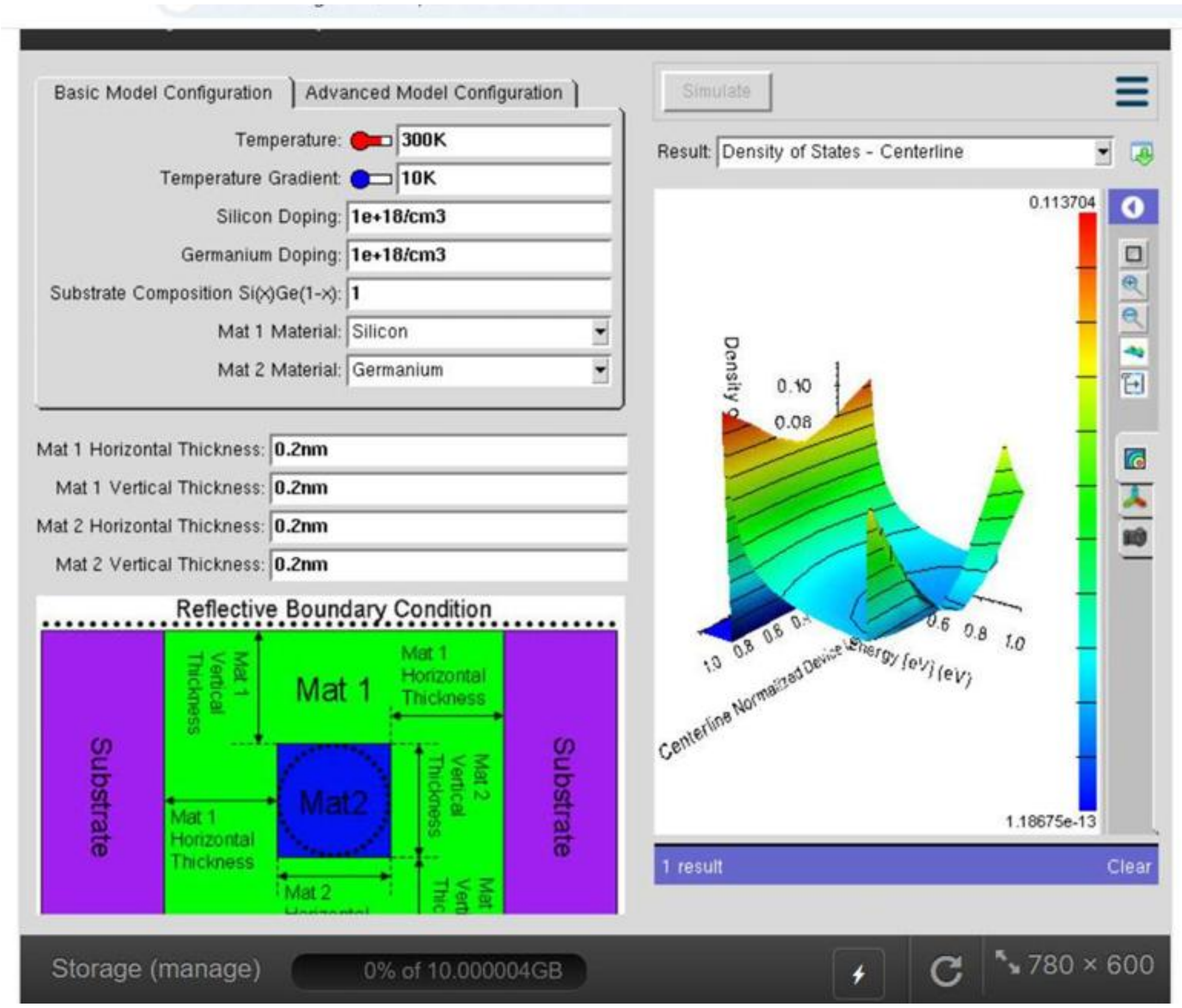
SLOT C-NANO ELECTRONICS

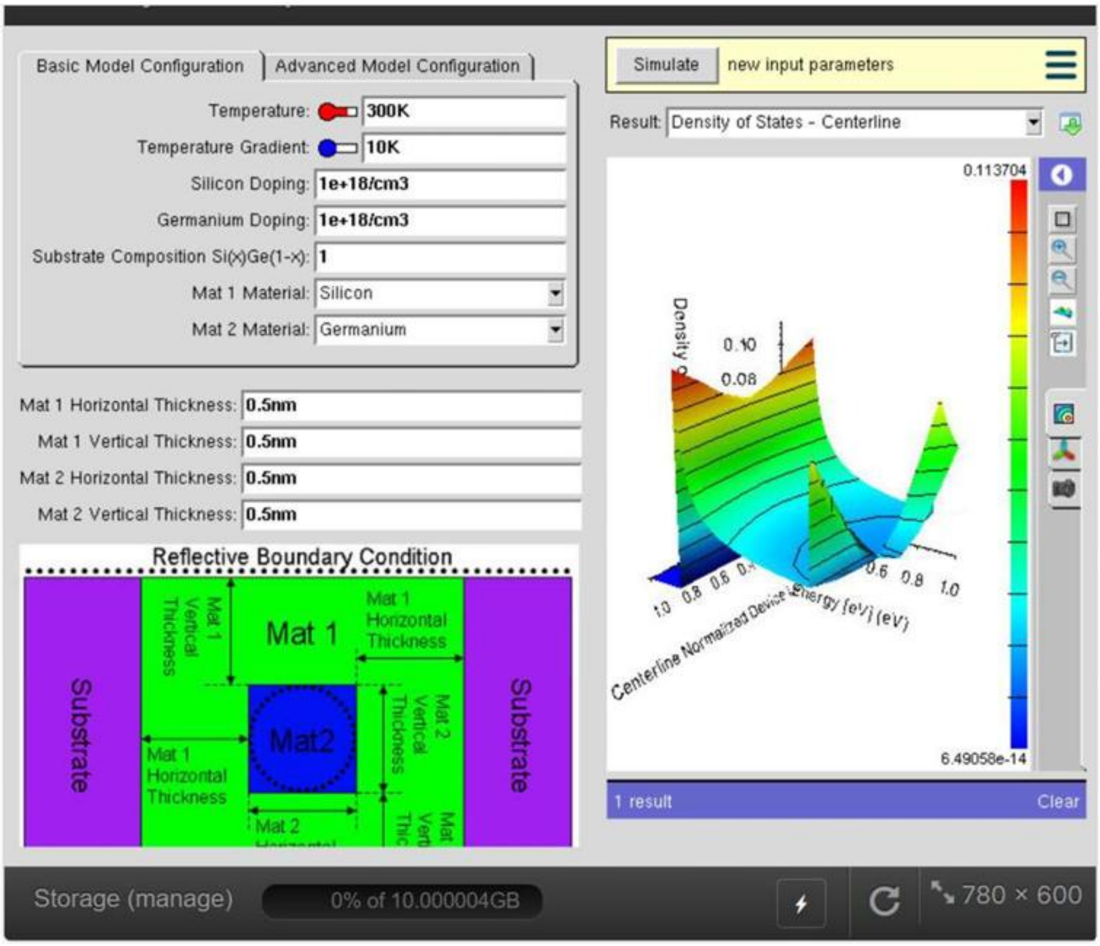
EXPERIMENT 2 - SIMULATION OF DENSITY OF STATED OF QUANTUM WELL STRUCTURE WITH VARIOUS NANOMETER THICKNESS

OUTPUT : Simulation of the density of states of quantum well of thickness 0.2 nm



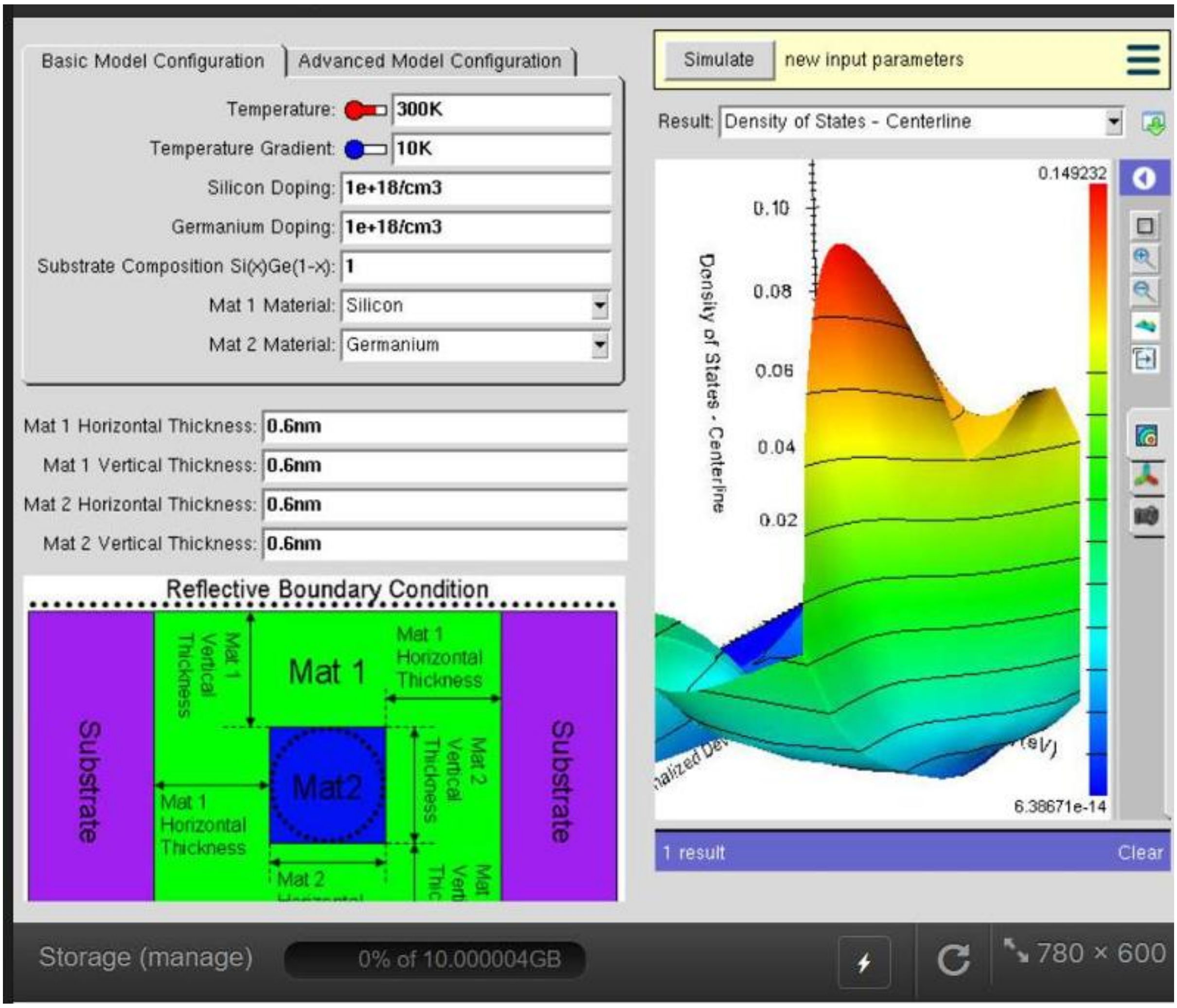
The output shows the density of states of a 0.2 nm quantum well. Strong quantum confinement creates discrete energy levels, producing a step-like DOS. As energy increases, more sub-bands contribute, increasing the density of states. This highlights the effect of ultra-thin quantum wells on electronic behavior.

CASE STUDY 1: simulate the density of states of quantum well of thickness 0.5nm



The simulation illustrates the density of states of a **0.5 nm quantum well**. Due to quantum confinement, the energy levels are quantized, producing a **step-like DOS**. The increased well thickness reduces confinement compared to thinner wells, resulting in more closely spaced energy levels.

CAE STUDY 2: simulate the density of states of quantum well of thickness 0.6 nm



Conclude:

The simulation shows that a 0.5 nm quantum well exhibits quantized energy levels and a step-like density of states due to quantum confinement. This demonstrates how quantum well thickness influences the electronic properties of semiconductor devices